

Software Process Simulation Modeling: Preliminary Results from an Updated Systematic Review

Chao Gao
Software Institute
Nanjing University
Nanjing 210093, P. R. China
mg1332004@software.
nju.edu.cn

Shu Jiang
Software Institute
Nanjing University
Nanjing 210093, P. R. China
js10@software.
nju.edu.cn

Guoping Rong
Software Institute
Nanjing University
Nanjing 210093, P. R. China
ronggp@software.
nju.edu.cn

ABSTRACT

Software Process Simulation Modeling (SPSM) has raised research interest since 1980s. However, it is observed that SPSM studies published in the ICSSP community may have dropped in recent years. The objective of this research is to update the recent status of this area. We conducted a Systematic Literature Review (SLR) using the QGS-based search strategy. The review identified 74 primary studies in the past five years (2008-2012). This paper presents the preliminary results from this updated SLR by answering the first four research questions. Based on the findings from this updated review, it can be concluded that in terms of the number of SPSM studies found in the overall software engineering community, there is no significant change (drop) compared to the former review stage (1998-2007).

Categories and Subject Descriptors

D.2.9 [Software Engineering]: Management—*Software process models*

General Terms

Management, Theory

Keywords

Software process simulation modeling, systematic literature review, QGS

1. INTRODUCTION

Software Process Simulation Modeling (SPSM) has become an increasingly active research area after being introduced by Abdel-Hamid and other pioneers [1] in 1980s. This approach provides effective tools to help evaluate and manage process changes made to software projects and organizations. ProSim workshops that focus on the state-of-the-art theories and applications of SPSM in addressing real-world problems started in 1998 and the first ProSim inaugurated a forum for SPSM research. In ProSim'98 Kellner, Madachy, and Raffo (KMR) published their highly-cited paper "Software process simulation modeling: Why? What? How?" [2] which discussed a variety of aspects of SPSM, such as

why to undertake SPSM, and simulation approaches/paradigms. Recently, Zhang et al. [3] researched the impact of process simulation on software practice.

In the community, Zhang et al. [4, 5, 7] carried out a systematic review which summarized relevant studies published during the decade (1998-2007) and presented the state-of-the-art of the SPSM research. Their work identified 156 studies on SPSM, 96 of which were published by ProSim/ICSP community. Liu et al. [6] reported an SLR on the role of SPSM in software risk management which is sub-scope of the former review.

ProSim was merged into ICSP in 2006, and ICSP was renamed to ICSSP in 2011. To understand recent progress of SPSM research, an updated SLR was carried out with the same research questions. In addition, the number of SPSM studies published by ProSim/ICSP community seemed to decline in recent years [8], an observation that also motivated our work to present an up-to-date and comprehensive view of SPSM in the Software Engineering (SE) community (not limited to ICSSP) of the past five years (2008-2012).

This paper, which is part of a synthetical research project, presents the preliminary results of an updated systematic review on SPSM. Section 2 describes the process of our SLR. The results and discussions of the review are presented in Section 3 and 4, followed by our conclusions in Section 5.

2. RESEARCH METHOD

Systematic literature review (SLR) is a "key research method to identify, evaluate, and interpret all available research relevant to a specific research question, or topic area, or phenomenon of interest" [9]. According to [10], SLR has become an important research methodology in SE.

The process of our SLR followed the Kitchenham's methodological guidelines for systematic review [9]. Different from the original steps introduced in the guideline, 'quasi-gold standard (QGS)' search strategy [11] which was a recently proposed research method for improving the validity and reliability of SLR was adopted in this inquiry. The review was mainly conducted by four researchers: two research students, one supervisor and one external checker. The two research students independently searched and selected primary studies, assessed the quality of primary studies, extracted data. The Kappa coefficient [12] was calculated to examine the inter-rater agreement between them on scales criteria. The supervisor played the role of expert panel who dealt with divergences between the two students during the process. Due to the limited space, this section only elaborates the distinctions from the previous stages of this SLR.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.

ICSSP'14, May 26–28, 2014, Nanjing, China
Copyright 2014 ACM 978-1-4503-2754-1/14/05...\$15.00
<http://dx.doi.org/10.1145/2600821.2600844>

2.1 Research Questions

The primary research question (RQ) of this systematic review is “What is the state-of-the-art of software process modeling simulation?” This question is decomposed into the following answerable research questions [4, 5]. Note that we only include and discuss the first four RQs in this report due to the page limit.

RQ1: What are the purposes or motivations of SPSM in the last 5 years’ practice? How are the purposes identified by KMR supported in SPSM research and practice?

RQ2: Which simulation paradigms have been applied in the last 5 years, and how popular are they in SPSM? Are there any new paradigms emerging?

RQ3: Which simulation tools are available for SPSM study and have been in use in the last 5 years? Which paradigm(s) does each of them support? How widely used are the tools?

RQ4: What are problem domains and modeling scopes have been the subjects of software process modeling?

2.2 Search Strategy

Even a well performed search process is unlikely to be able to collect papers completely [13]. In this review, the approach of ‘quasi-gold standard’ (QGS) proposed by Zhang et al.[11] was adopted, which is based on a set of known studies gathered by manual search within certain venues and fixed time span, to objectively define the search string and evaluate the performance of the search string.

2.2.1 Selection Criteria

Different from the guidelines [9], the selection criteria were identified from the existing criteria for SLR on SPSM [4, 5] before defining the search string. The inclusion and exclusion criteria are presented in Table 1. The following criteria were applied in both manual and automated search stages.

Table 1. Inclusion/Exclusion criteria

<i>Inclusion Criteria</i>	<i>Exclusion Criteria</i>
C1: Published time between 2008.1-2012.12	C3: In the form of books
C2: Focus on the primary studies employing simulation modeling paradigm(s) for software process research, education and practice	C4: In the form of editorial, abstract, keynote, poster or a short paper (body less than 5 pages)
	C5: Opinion pieces, comments, corrections, notes, slides alone or position papers
	C6: Focus on summarizing the outcomes of the existing research work, e.g. roadmap, review, survey, etc.
Papers meet C1 AND C2 will be included.	
Papers meet C3 OR C4 OR C5 OR C6 will be excluded.	

In addition to these inclusion/exclusion criteria, we excluded the duplicate papers and selected repeated papers at each stage of selection. Duplicated papers mean the identical paper but indexed by several digital libraries. Among duplicates, only the one from the publisher’s database was kept. Repeated papers refer to one study published in different venues with different abstract but similar content. The most recent and comprehensive version or continued studies of the repeated papers were kept at the end of the search step.

2.2.2 Manual Search

For the purpose of producing or validating a search string for automated searching, ProSim/ICSP related publication channels were first searched manually. The sources of manual search are summarized in Table 2. After manual search, the Kappa value between two students reached 0.85 which meant the results

between them were highly consistent according to [11]. After discussion, the differences between the students were eliminated. Table 3 shows the study distribution across the venues. Note that five repeated studies with different versions were both counted in ICSP and SPIP/JSEP. After the manual search, 37 primary studies were identified.

Table 2. Sources of literature search

<i>Resource</i>	<i>Period</i>	<i>Search method</i>
Conference proceedings		
ICSE	2008 - 2012	Manual
PROFES	2008 - 2012	Manual
ESEM	2008 - 2012	Manual
SEKE	2008 - 2012	Manual
ICSP/ICSSP	2008 - 2012	Manual
Journals		
TSE	2008 - 2012	Manual
TOSEM	2008 - 2012	Manual
JSS	2008 - 2012	Manual
SPIP/JSME/JSEP ¹	2008 - 2012	Manual
JSS	2008 - 2012	Manual
IST	2008 - 2012	Manual
IJSEKE	2008 - 2012	Manual
Digital libraries		
IEEE Xplore	2008 - 2012	Automated
ACM digital library	2008 - 2012	Automated
Springer Link	2008 - 2012	Automated
ScienceDirect	2008 - 2012	Automated
Wiley Online Library	2008 - 2012	Automated

Table 3. Frequency of papers in QGS

<i>Conference</i>	<i>Frequency</i>	<i>%</i>	<i>Journal</i>	<i>Frequency</i>	<i>%</i>
ICSP/ICSSP	25	92.6	JSS	2	20
ESEM	1	3.7	IST	2	20
ICSE	0	0	SPIP/JSME/JSEP	6	60
PROFES	1	3.7	TSE	0	0
SEKE	0	0	TOSEM	0	0
—	—	—	IJSEKE	0	0
Total	27	100	Total	10	100

2.2.3 Defining Search String

The search string was validated with high quasi-sensitivity in the former review stage (1998-2007) [5]. The same string was evaluated again for its effectiveness during the period of this update (2008-2012). The following search string was coded into equivalent forms to match the syntax of each digital library in Table 2, and then the quasi-sensitivity was calculated to evaluate the string and adjust it if needed.

((software process) OR (software project) OR (software product) OR (software evolution)) AND (simulation OR simulator OR simulate OR (dynamic model) OR (system dynamics))

Quasi-sensitivity is an important criterion for evaluating the quality and efficiency of a search strategy [11]. It refers to the proportion of relevant studies covered by the QGS and can be calculated as:

$$\text{quasi-sensitivity} = \frac{\text{Number of retrieved relevant studies covered by QGS}}{\text{Total number of relevant studies in QGS}} \cdot 100\%$$

In the evaluation of the above search query, merely one study was missed. The corresponding quasi-sensitivity reaches 97.3% (36/37), which means a high precision according to [11]. It confirms the search string was still valid for carrying out automated search in the updated SLR.

¹ SPIP and JSME were merged into JSEP in 2012.

2.2.4 Automated Search

After validating the search string, an automated search was carried out. The Kappa value reached 0.88, which shows the study selections of the two research students were remarkably consistent. Any disagreements were eliminated through discussions with the supervisor. Table 5 presents the results of the automated search.

Table 5. Result of automated search and QGS (manual)

Search Engine	# of Retrieved studies	# of Relevant studies	# in QGS (manual)
IEEE Xplore	96	17	4
ACM DL	19	6	2
ScienceDirect	19	1	4
SpringerLink	1825	17	21
Wiley	386	3	6

The total of relevant studies by automated only : 44

The total of relevant studies by manual only : 37

The total of all relevant studies after removing repeats in SLR : 74

2.3 Study Classification

In terms of the experience of the previous review, four study categories were identified which were effectively used in study classification in [4, 5]. They were also applied in our review stage:

- A: Software process simulation models or simulators;
- B: Process simulation modeling paradigms, methodologies, and environments;
- C: Applications, guidelines, and solutions for adopting process simulation in software engineering practice;
- D: Experience reports of SPSM research and practice.

These four types of studies focus on different aspects of software process simulation research, and may give the answers from different points of view to different research questions. The relationships among them were discussed in [7] and more details of the classification in previous stages can be found in [4, 5].

2.4 Quality Assessment and Data Extraction

Study quality was assessed using a set of questions, some common to the categories and other specific to each category, that were defined in [4, 5]. For each study, each applicable question was answered with a value of 1 (yes), 0.5 (partial), or 0 (no). Space limit prohibits listing questions and scores here, but the extracted attributes can be found in [4, 5].

3. RESULTS

After the thorough search and careful selection, 74 primary studies on SPSM were identified from 1998 to 2012. They form an addition of the comprehensive body of SPSM research and practice for the past five years. In terms of the number of the identified primary studies, it can be observed that there is no significant change (drop) compared to the average number of the studies per year in [4, 5] between 1998 and 2007. The full list of the primary studies identified is available upon request.

Figure 1 shows the number of SPSM studies published during the past 15 years grouped by ProSim/ICSSP community and the general SE community respectively. Compared to the former review stage (1998-2007), we find that the number of studies published in ProSim community shows a decreasing trend in the past five years (2008-2012). In contrast, the overall number of studies published in SE per year was oscillating between 10 and 20. The primary studies published in conference/workshop were more than the studies published in journals either in each year or over the entire period (58 vs. 16).

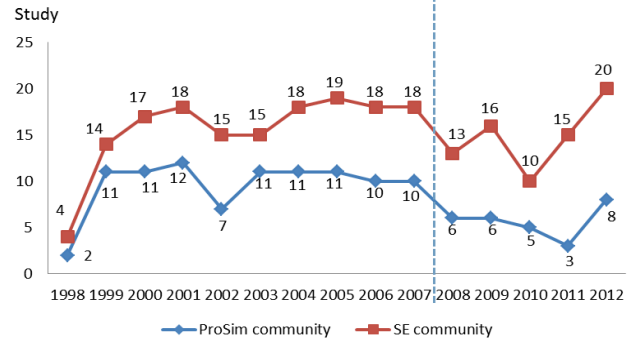


Figure 1. Number of publications by community

3.1 Classification

According to Zhang et al. [4, 5] and the pilot review of our work, four study categories were identified. By reviewing the full text of the papers, all primary studies were classified into at least one category (A, B, C, and D). The distribution of studies per category and year are shown in Figure 2.

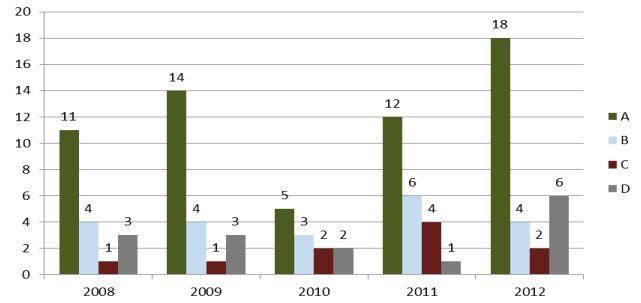
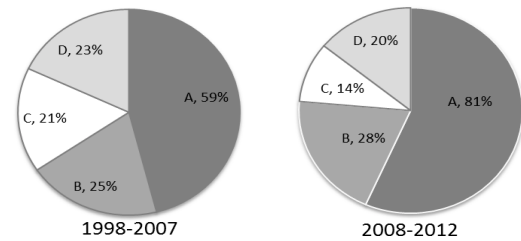


Figure 2. Summary of study categories distribution

As shown in Figure 2, most primary studies were identified as Category A for both the total (81%) and each year separately. In sum, there have been 57 software process simulation models (simulators) developed and published in various proceedings and journals during the past five years. Only 10 primary studies (14%) were identified as Category C, while 28% of primary studies were of Category B and 20% of Category D. 26 studies were classified into two categories, and 3 studies were identified as combinations of three categories.



Because some studies were classified into multiple categories, the sum of the pie chart is a bit over 100%

Figure 3. Study classification in the different stage

Figure 3 compares the distributions of studies over the four categories between the two review stages. The occupation of A in our review stage (2008-2012) has an increase of 22% compared to the former stage (1998-2007) [5], while a 7% decrease of C.

3.2 Quality

The primary studies were assessed for quality using the checklist in [4]. Although the full score for each study cannot be included in this paper, we present the normalized average quality score per source type (proceedings or journals) and year in Figure 4. The study quality of conference/workshop proceedings has a gradual upward from 2009 to 2012. In most cases, the quality of journal articles was better than of the proceedings papers of the same year except in 2010. Overall, many studies of Category A failed to make a trial of the model on an industry scale problem. For Category B and C studies, consistent with the fact found in [4], the limitations associated with the paradigm/method/solution were still rarely discussed.

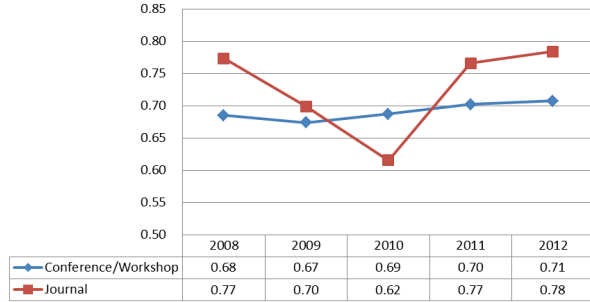


Figure 4. Average study quality per source type and year

4. DISCUSSIONS

We discuss the preliminary findings to the four research questions (Q1 to Q4) defined in Section 2.1 in this section.

4.1 Purposes for SPSM (Q1)

Zhang et al. [4, 5] extended and restructured KMR's six specific purposes [2] for SPSM to be ten purposes based on the observation of SPSM studies. This update of the SLR confirmed that these ten purposes for SPSM research identified in [4]. These research purposes were grouped in three levels (i.e. cognitive level, tactical level, strategic level). They were further detailed as 10 purposes and the details can be found in [4].

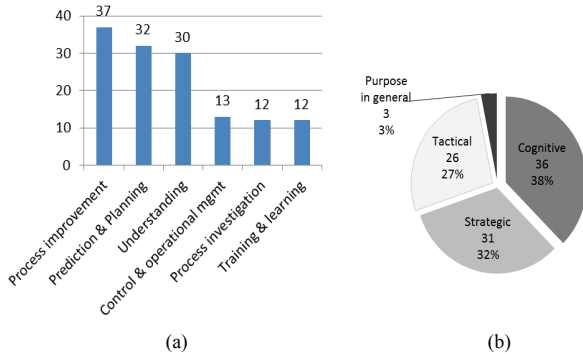


Figure 5. Study distribution by research purpose

Figure 5 shows the study distribution by research purpose in this review stage. The frequently researched purposes which more than 10 studies were identified into were shown in (a). Figure 5-b denotes the cognitive purpose dominates SPSM studies (38%). The sum of the four sectors in (b) is larger than the number of primary studies because some studies were counted for both cognitive and strategic purposes.

4.2 Modeling Paradigms (Q2)

In our review, 11 simulation modeling paradigms were found. Figure 6 shows the paradigms with more than one study applied. As before, System Dynamics (SD, 45%) and Discrete-Event Simulation (DES, 33%) were still the most widely used techniques in SPSM in the past five years. Other paradigms included Agent-Based Simulation (ABS, 14%), Qualitative (semi-quantitative) Simulation (QSIM) and Petri-Net Simulation.

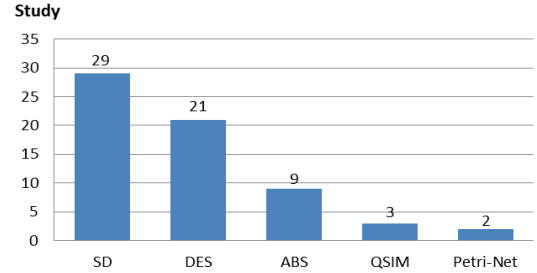


Figure 6. Study distribution by simulation paradigms

ABS paradigm has a rising trend compared with the data reported in [3]. QSIM still accounted for a small proportion. In addition, other paradigms included state-based simulation (SBS), role-playing game and discrete-time simulation (DTS) which appeared in the previous period were not applied in this review period.

During the past five years, hybrid simulation has still been one of the most frequent research themes in SPSM. Most of these studies (7 papers) presenting the hybrid simulation continue to focus on the combination of continuous (SD) and discrete-event simulation (DES).

4.3 Simulation Tools (Q3)

Overall, 13 tools were identified with respect to the simulation models/simulators presented in Category A studies. Their application numbers are shown in Table 7. The total application number (26) is less than the number of Category A studies (60) as some studies did not explicitly name their simulation tools.

Table 7. Application of simulation tools

Paradigm	Simulation tool or package
Hybrid (SD+DES)	Extend(1), MathModelica(1), iThink(1), EPF(1), AnyLogic(1)
SD	Vensim(5), iThink(3), EPF(1), Simulink(1), Illium(1), VPMsim1.0(1), matlab(1)
DES	Extend(2), SimJava(1), AnyLogic(1)
ABS	Anylogic(1)
QSIM	Vensim(1)
General	Crystall Ball(1), Kermeta(1)

There have been many new simulation tools emerging during this period, however, Vensim, iThink, and Extend which were found as the top three applied tools in [4] still keep their leading positions. The use of Vensim exceeded the other tools, which is consistent with the phenomenon that VensimTM (by Ventana Systems, Inc. www.vensim.com) is the most popular tool for continuous simulation that many simulation models were based on. ExtendTM (by ImagineThat, Inc. www.extendsim.com) has again been the first choice by the discrete-event and hybrid modelers in SPSM community since it is able to support both continuous and discrete simulation models. On the other side, many simulation tools found in [4] disappeared in this review such as Repast, Netlogo, DSOL, etc. AnyLogicTM (by XJ Technologies, Inc. www.anylogic.com) is the only simulation tool that supports DES, ABS, and SD simultaneously. It replaces Netlogo and becomes the first choice by ABS modelers.

4.4 Problem Domains and Model Scopes (Q4)

In total, 14 problem domains were extracted in this review from Category A studies. “Management” studies model and simulate the process management or the project management.

We refined the six scopes with reference to [4, 5], i.e. single-phase, multi-phase, project, program or portfolio, evolution, and long-term organization, when reviewing simulation model scope. Table 8 shows the relations between the modeling domains and scopes extracted from Category A studies. The number in the ‘subtotal’ row is not always the exact sum of the column. This is because some studies modeled multiple domains, e.g. a combination of software evolution and software/program verification. ‘Project’ was the most frequently modeled study scope, particularly for ‘Management’. ‘Single-phase’ and ‘Program or Portfolio’ were the next two most studied topics. “Management” has been the most frequently modeled topic among these domains, while most of them modeled software process management.

Table 8. Modeling problem domains vs. model scopes

	Single-phase	Multi-phase	Project	Program or Portfolio	Evolution	Long-term organization	Unknown or N/A	Sum
Management	4	1	16	3	1	1	2	28
SE process			5	1				6
Software evolution				1	3			4
Requirement/specifications			3				1	4
General			2				1	3
Design tools & techniques		1	1					2
Distribution, Maintenance & Enhancement			2					2
Human factors in software design			2					2
Metric/Measurement	1	1						2
Software product lines		1			1			2
Software/program verification	1			1				2
Global software engineering			1					1
Software quality/SQA			1					1
Testing & Debugging		1						1
Subtotal	6	4	33	5	4	1	4	60

5. CONCLUSIONS

We conducted an updated systematic literature review of software process simulation modeling by systematically searching and aggregating studies published from 2008 to 2012. As a continued study of Zhang et al. [4, 5, 7], this paper presents the up-to-date results and reports the preliminary findings from the extended review stage. The results of our review indicate that: (1) in terms of the average number of studies found in SE community, there is no significant change compared to the former review period though the number of the studies published in ProSim/ICSSP community declined; (2) many Category A studies lack trials of their proposed models on an industry scale problems; (3) due to the enhanced capability in capturing the characteristics of complex real-world software processes, hybrid process simulation models have still attracted great interest.

A more detailed analysis of the studies of Categories B, C and D and an investigation of the issues (Q5 and Q6) raised in [4] but not reported in this paper will be included in our future work.

6. ACKNOWLEDGMENTS

We would like to record our special appreciation to Professor He (Jason) Zhang (Nanjing University) for his encouragement, guidance, suggestions, and review of this work.

7. REFERENCES

- [1] Abdel-Hamid, T., & Madnick, S. E. 1991. *Software project dynamics: an integrated approach*. Prentice-Hall.
- [2] Kellner, M. I., Madachy, R. J., & Raffo, D. M. 1999. Software process simulation modeling: why?what? how?. *Journal of Systems and Software*. 46, 2 (Apr. 1999), 91-105.
- [3] Zhang, H., Jeffery, R., Houston, D., Huang, L., & Zhu, L. 2011. Impact of process simulation on software practice: an initial report. In *Proceedings of the 33rd international conference on Software engineering*. ACM, New York, NY, 1046-1056.
- [4] Zhang, H., Kitchenham, B., & Pfahl, D. 2008. Reflections on 10 years of software process simulation modeling: A systematic review. In *Making globally distributed software development a success story*. Springer Berlin Heidelberg, 345-356.
- [5] Zhang, H., Kitchenham, B., & Pfahl, D. 2010. Software process simulation modeling: an extended systematic review. In *New Modeling Concepts for Today's Software Processes*. Springer Berlin Heidelberg, 309-320.
- [6] Liu, D., Wang, Q., & Xiao, J. 2009. The role of software process simulation modeling in software risk management: A systematic review. In *Empirical Software Engineering and Measurement, 2009. ESEM 2009. 3rd International Symposium on*. IEEE, 302-311.
- [7] Zhang, H., Kitchenham, B., & Pfahl, D. 2008. Software process simulation modeling: Facts, trends and directions. In *Software Engineering Conference, 2008. APSEC'08. 15th Asia-Pacific*. IEEE, 59-66.
- [8] Zhang, H. 2012. Special Panel: Software Process Simulation—At a Crossroads?. In *Software and System Process (ICSSP), 2012 International Conference on*. IEEE, 215-216.
- [9] Kitchenham, B., Charters, S. 2007. Guidelines for performing systematic literature reviews in software engineering.
- [10] Zhang, H., & Babar, M. A. 2011. An empirical investigation of systematic reviews in software engineering. In *Empirical Software Engineering and Measurement (ESEM), 2011 International Symposium on*. IEEE, 87-96.
- [11] Zhang, H., Babar, M. A., & Tell, P. 2011. Identifying relevant studies in software engineering. *Information and Software Technology*, 53, 6 (Jun. 2011), 625-637.
- [12] Cohen, J. 1968. Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit. *Psychological bulletin*, 70, 4 (Oct. 1968), 213-220.
- [13] B.A., Kitchenham, Z., Li, and A., Burn. 2011. Validating Search Processes in Systematic Literature Reviews. In *Proceedings of EAST*.